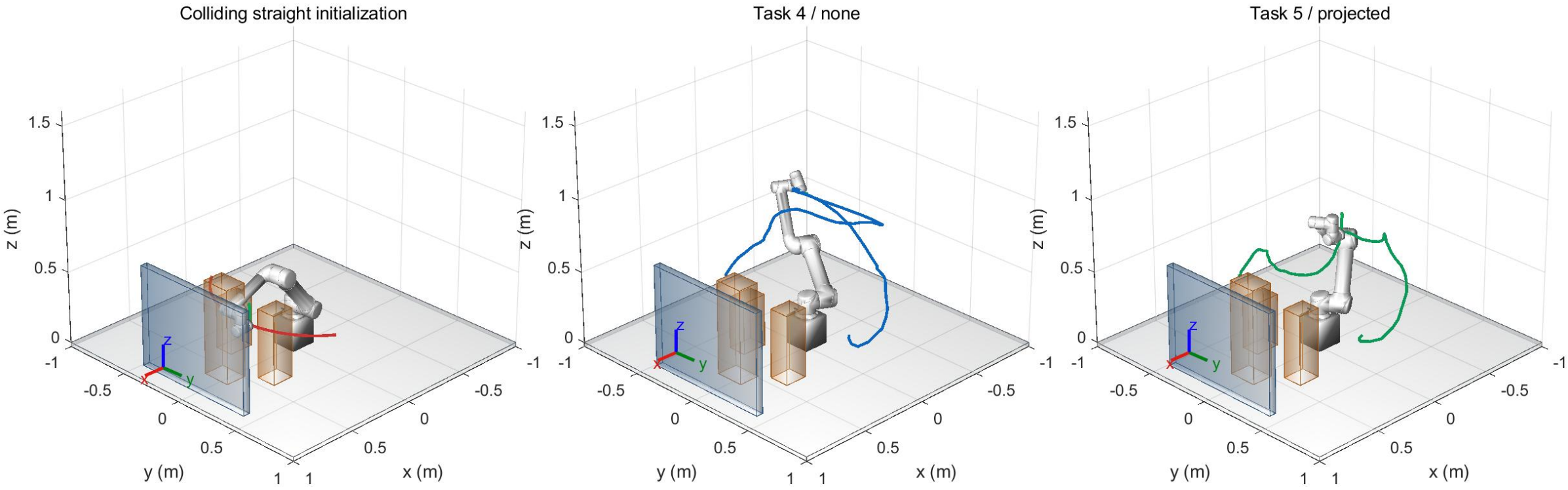
Task 4: a mounted UR5 workcell



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initial states collide

Same endpoints and straight initialization for both tasks

Task 5: upright tool direction

Tool y aligns with world +z

Yaw remains free

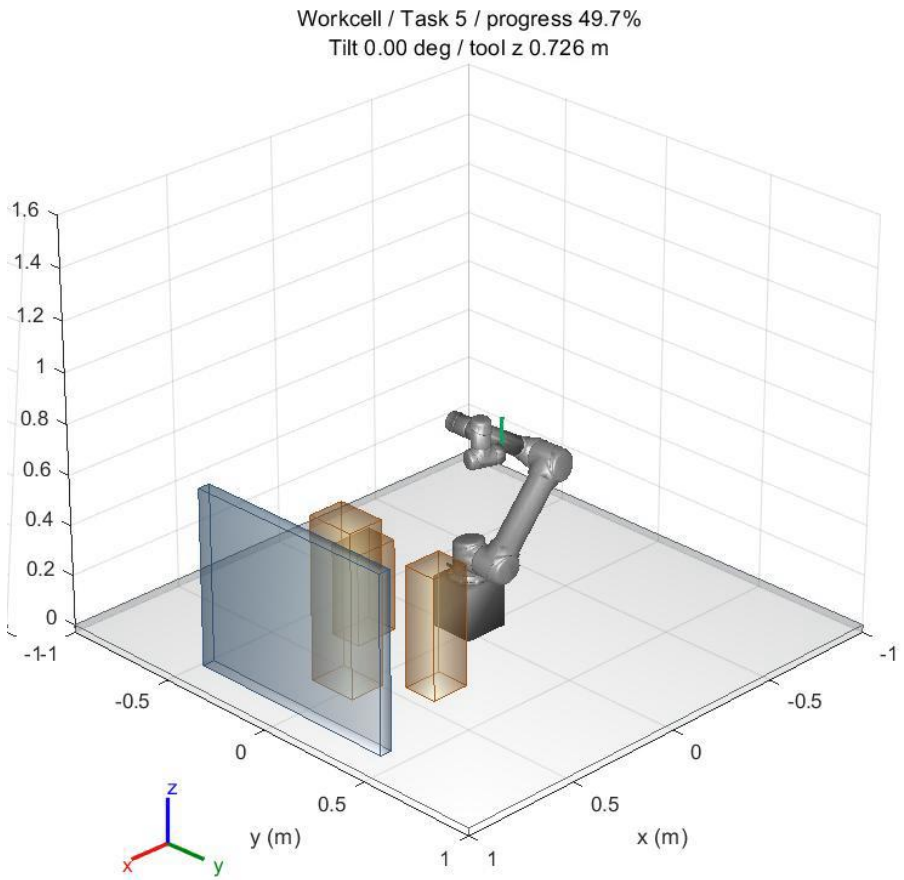
Direction penalty, lambda = 1000

Damped Jacobian projection

Self-collision cost and rejection

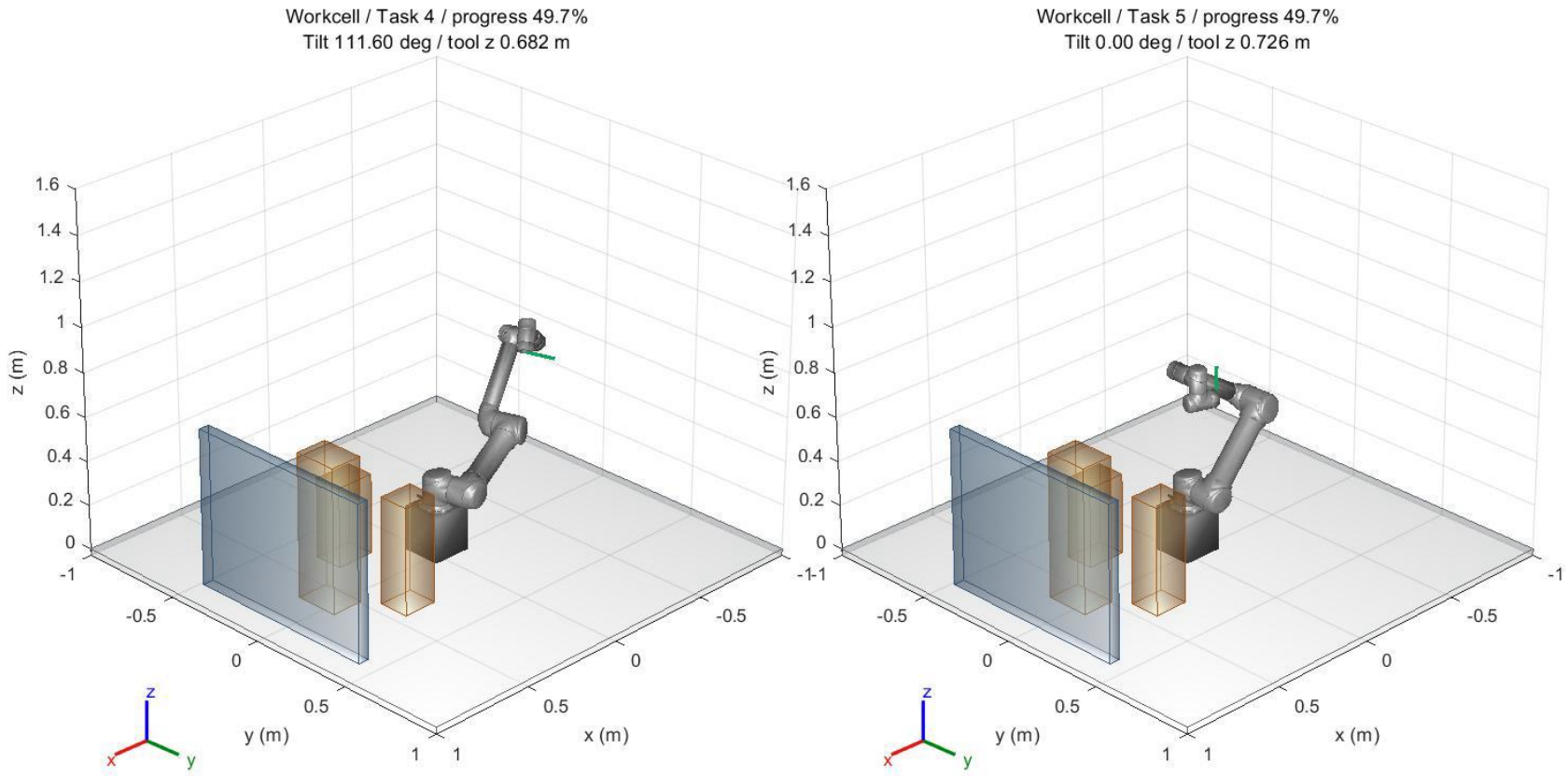
Maximum accepted tilt: 5 degrees

Conservative clearance: 35 mm



Click video to play / 10 s

Same problem, different tool orientation



Demo maximum tilt: 120.29 degrees (Task 4) / 0.930 degrees (Task 5)

Seed 5300. Normalized path progress, without an execution-time model.

Held-out results and contribution

Condition	Success	Median time (s)	Worst tilt (deg)	Min proxy gap (mm)
Task 4	20/20	30.42	169.591	35.41
Task 5	20/20	68.19	0.997	35.23

20 paired seeds. Every successful path also passes the 0.00625 rad check.

My contribution: scene, upright projection, self-collision handling, paired experiments and independent verification.

V4/V5 extensions in backup slides. Discrete geometry checks; no payload dynamics.